RESCUE JUNIOR - Bill of Materials (BOM)

Team name: simple

Every Line/Maze team has to submit a bill of materials for their robot

Instructions:

*Enter Team name above

*All costs need to be in local currency and their approximate conversion into dollars.

*All components worth less can be summarized in one line quantity 1 and their total cost (e.g. Screws)

*In the Local Currency Name column, the team must enter the name of the local currency, if it is different from US dollars.

*In the software sheet, in the Author column, the name of the designer, writer, or company that owns the software or library should be entered. If the team created their own algorithm, dataset, or AI model, they can include their names as authors.

*On the hardware sheet, in the "Author" column, enter the name of the company, engineer, or part manufacturer. If the team made a custom part, such as the robot housing, the team should include their names as the authors.

IMPORTANT The "Hardware" sheet is not required for simulation

Team name: simple

Name of the Local Currency	Total Cost Local Currency	Total Cost ($) U.S.A. Dollars
Brazilian Real	R$11,824.09	US$2,094.00

#	Component	Part name	Autor	Source	Quantity	Unit cost (Local Currency)	Unit cost (USA Dollars)	Total Cost (Local Currency)	Total Cost (USA Dollars)
1	PCB with electronic components	Custom made	Name of team members/Jlcpcb	jlcpcb.com	1	R$338.80	US$60.00	R$338.80	US$60.00
2	Wheels	Nexus Robot Mecanum Wheel	Nexus Robot	robotshop.com	4	R$112.93	US$20.00	R$451.72	US$80.00
3	Sheet metall parts for frame	Custom made		Local workshop	1	R$1,129.34	US$200.00	R$1,129.34	US$200.00
4	Drive motors	Maxon ECX TORQUE 22 XL Ø22 mm	Maxon Group	Maxon Motors	4	R$1,694.01	US$300.00	R$6,776.04	US$1,200.00
5	Robots shell	Custom made	Name of team members	3D Printer	1	R$112.93	US$20.00	R$112.93	US$20.00
6	Robots costume	Custom made	Name of team members	Self made	1	R$282.34	US$50.00	R$282.34	US$50.00
7	Microcontroller	Arduino Mega 2560	Arduino	conrad.de	2	R$237.16	US$42.00	R$474.32	US$84.00
8	Connecting parts	Screws, nuts, washers, ...		Local tool store	1	R$225.87	US$40.00	R$225.87	US$40.00
9	Motors for arm	NEMA-17-01 NEMA-17-01 0.4 Nm 1.68 A	Conrad Electronic	conrad.de	2	R$118.58	US$21.00	R$237.16	US$42.00
10	Stepper motor controllers	Pololu Tic T825	Pololu Robotics and Electronics	conrad.de	2	R$508.20	US$90.00	R$1,016.40	US$180.00
11	LEDs	Various colours and sizes	Farnell An Avnet Company	farnell.com	20	R$2.82	US$0.50	R$56.40	US$10.00
12	Battery	11.1 V 3700 mAh	Farnell An Avnet Company	farnell.com	3	R$203.28	US$36.00	R$609.84	US$108.00
13	Cables	Various cables		Local electronics store	1	R$112.93	US$20.00	R$112.93	US$20.00
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Team name:	simple

Name of the Local Currency	Total Cost Local Currency	Total Cost ($) U.S.A. Dollars
Brazilian Real	R$0.00	US$0.00

#	Name	Software's Tool/Library	Description	Source	Autor	Cost (Local	Cost (USA
1	VL53L0X-Arduino	Library	For controlling the VL53L0X distance sensor on Arduino	pololu/vl53l0x-arduino: Biblioteca Po	Pololu	FREE	FREE
2	Open CV	Library	Image processing library	OpenCV - Biblioteca abierta de Com	Opencv.org	FREE	FREE
3	IDE Arduino	Software's Tool	Software	Software | Arduino	Arduino.cc	FREE	FREE
4	Visual Studio Code	Software's Tool	Software	Visual Studio Code: edición de códi	Microsoft	FREE	FREE
5	Numpy	Library	Array and math library	NumPy	Numpy developers	FREE	FREE
6	Matplotlib	Library	Python graphics generation library	Matplotlib — Visualización con Pyth	Matplot lib dev team	FREE	FREE
7	Tinckercad	Software's Tool	Online platform for simulating circuits	Tinkercad	Autodesk	FREE	FREE
8	Git	Software's Tool	Version control for managing functions	Git - Descargas	Software Freedom Conservancy	FREE	FREE
9	Fusion 360	Software's Tool	3D CAD shell design	Fusión de Autodesk | Software basa	Autodesk	FREE/EDUCATION	FREE/EDUCATION
10	Tic Arduino	Library	Library for controlling a Nema 17 stepper motor with the Tic T825 driver	https://github.com/pololu/tic-arduino	DavidEGrayson	FREE	FREE
11	Modulemotorl298n	Library	The library is designed to make it easy to use motors connected to an L298N module		Name of team members	FREE	FREE
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Team name:	simple

Name of the Local Currency	Total Cost Local Currency	Total Cost ($) U.S.A. Dollars
Korean Won KRW	1,321,378	US$871.71

FX KRW per	1,515.85
FX KRW per	1,763.26
Notes	Notes
	USD/KRW a
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#	Component	Part name	Autor	Source	Quantity	Unit cost	Unit cost (US Dollars)	Total Cost (Local	Total Cost	
1	Controller hub	LEGO Technic Large Hub for SPIKE Prime	LEGO Education	https://	1	530,472	349.95	530,472	349.95	Approx.
2	Vision controller	OpenMV Cam H7 R2	OpenMV Official Store	Order	2	99,750	65.80	199,500	131.61	Final plan:
3	Camera/Hub interface board	SPIKE Smart Camera Breakout Board (spike-	Anton's Mindstorms	Website	1	47,608	31.41	47,608	31.41	€27.00 ×
4	Distance sensor	VL53L4CD ToF module	RTRobot / AliExpress	Order	3	13,655	9.01	40,965	27.02	
5	I2C multiplexer	TCA9548A 1-to-8 I2C expansion board	RTRobot / AliExpress	Order	1	8,990	5.93	8,990	5.93	
6	Touch sensor	Micro limit switch	TLZWLA	Order	2	979	0.65	1,958	1.29	
7	Lens	1.8mm M12 wide-angle lens (IRC650 filter)	Shop1103340668 /	Order	2	5,200	3.43	10,400	6.86	
8	Camera extension cable	OpenMV camera module extension cable	OpenMV Official Store	Order	1	21,757	14.35	21,757	14.35	
9	Utility servo	Geekservo programmable 2kg motor (blue)	Wishiot Store	Order	1	9,360	6.17	9,360	6.17	
10	Power bank	Xiaomi Ultra Slim 5000mAh	Xiaomi official partner	Order	1	74,800	49.35	74,800	49.35	
11	Drive motor	LEGO Technic Large Angular Motor (45602)	LEGO Education	https://	4	84,812	55.95	339,248	223.80	Approx.
12	Wheel / tire	LEGO Tire 81.6 × 38 R Balloon, part 45982	LEGO / ATLBrick	https://	4	9,080	5.99	36,320	23.96	Approx.
14	Custom fabricated parts	3D-printed brackets / mounts designed for the	Team simple	User-	1					Designed in
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FX KRW per 1,515.85

Team name: simple

Name of the Local Currency	Total Cost Local Currency	Total Cost ($) U.S.A. Dollars
Korean Won	0	US$0.00

#	Name	Software's Tool/Library	Description	Source	Autor	Cost (Local	Cost (USA
1	Main robot firmware	tof_circle.py / tof_circle_ru	OpenMV circle + 3x VL53L4CD + PUPRemo	Local project files	Team / custom	0.00	0.00
2	Prime hub control	Pybricks program	SPIKE Prime driving, mission logic, camera v	pybricks.com	Pybricks contributors	0.00	0.00
3	OpenMV camera developmen	OpenMV IDE	OpenMV camera programming, deployment,	https://openmv.io/pages/download	OpenMV	0.00	0.00
4	Communication library	PUPRemote	Hub-camera communication layer	Project library / uploaded files	PUPRemote	0.00	0.00
5	Distance sensor library	vl53l4cd.py	VL53L4CD distance sensor driver used by O	Local project files	ST-based port /	0.00	0.00
6	Runtime / scheduler	MicroPython + uasyncio	OpenMV runtime and async task scheduling	micropython.org	MicroPython contribut	0.00	0.00
7	Code editor	Visual Studio Code	Editing and managing Prime / MicroPython p	https://code.visualstudio.com/download	Microsoft	0.00	0.00
8	3D CAD design	Autodesk Fusion (formerly	Designing custom brackets and parts for 3D	https://www.autodesk.com/education/edu-	Autodesk	0.00	0.00
9	3D-print preparation	Bambu Studio (usage to c	Slicing and preparing custom parts for Bamb	https://bambulab.com/en/download/studio	Bambu Lab	0.00	0.00
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